AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

Student Feedback & Teaching Improvement Analyzer

[AI-Driven Teaching Excellence Analyzer]

Project Domain

[Excel, Power BI, Copilot/Chatbot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

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Abstract

This project, *Student Feedback & Teaching Improvement Analyzer (AI-Driven Teaching Excellence Analyzer)*, was undertaken as part of the **Artificial Intelligence in the Contemporary World (AICW) Fellowship Program**, supported by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP.

The project addresses the challenge of transforming student feedback into actionable insights for teaching improvement. The objective was to design an AI-based system that analyzes feedback data and provides clear guidance to help educators enhance their methods.

Using **Microsoft Excel, Power BI, and Copilot assistance**, we performed data cleaning, processing, and visualization. Dashboards were developed to highlight teaching clarity, engagement, and overall satisfaction. The analysis revealed patterns that identified both strengths and areas requiring improvement.

In conclusion, this project demonstrates how AI and analytics can convert simple feedback into meaningful guidance, supporting continuous teaching excellence and improved student outcomes.

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Chapter 1: Introduction

This chapter introduces the project undertaken as part of the **Artificial Intelligence in the Contemporary World (AICW) Fellowship Program**, organized by Edunet Foundation in collaboration with Microsoft, LinkedIn, and SAP. The fellowship provides students with practical exposure to Artificial Intelligence, Data Analytics, and Business Intelligence tools through hands-on projects.

Our project, *Student Feedback & Teaching Improvement Analyzer (AI-Driven Teaching Excellence Analyzer)*, is designed to address a common challenge in education: while student feedback is collected regularly, it is often not analyzed in a way that leads to meaningful teaching improvement. As a result, valuable insights about teaching clarity, engagement, and overall satisfaction remain unused.

The motivation behind this project is to make feedback more useful for both teachers and institutions. By applying AI-driven techniques, feedback can be transformed into clear, actionable insights that highlight strengths and areas needing attention. This helps educators refine their teaching methods and supports institutions in achieving continuous improvement.

The fellowship projects span multiple domains such as **Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting**. Within this broad framework, our project contributes to the education domain by showing how technology can enhance teaching practices and improve student learning outcomes.

1.1 Problem Statement

Despite the current system's ability to analyse student feedback, several **key limitations remain unaddressed**, restricting its effectiveness and innovation potential. These gaps highlight opportunities for developing a more **robust, intelligent, and award-worthy solution**:

1. Lack of Multilingual Feedback Processing

The existing system does not support feedback submitted in **regional or native languages**, leading to the exclusion of a significant portion of student responses. This results in a **representation bias** and limits the inclusivity and comprehensiveness of the analysis. There is a need for **multilingual NLP capabilities** to ensure equitable consideration of all student voices.

2. Absence of Longitudinal Performance Tracking

The current design focuses on **single-instance analysis** and lacks mechanisms to track and compare faculty performance over **multiple semesters or academic years**. Without longitudinal insights, it becomes difficult to identify **trends, progress, or decline**, thereby limiting strategic academic planning and continuous improvement.

3. Unaddressed Bias in Anonymous Feedback

Anonymous feedback systems are inherently vulnerable to **biases such as exaggerated praise, personal grievances, or irrelevant/spam responses**. The absence of bias detection and filtering mechanisms can

distort sentiment analysis, leading to **misleading conclusions and unfair evaluations**. A more intelligent system must incorporate **bias detection and credibility assessment models**.

4. Lack of Personalized Faculty Improvement Roadmaps

While the system identifies strengths and weaknesses, it does not extend to generating **actionable, personalized development plans** for educators. There is a gap in translating insights into **tailored recommendations, training resources, or skill enhancement pathways**, which are essential for meaningful teaching improvement.

5. Missing Real-Time Feedback and Intervention Mechanism

The current approach processes feedback only **after course completion**, which delays corrective action. This post-facto analysis fails to leverage the opportunity for **mid-semester intervention**, where timely insights could significantly enhance ongoing teaching effectiveness and student learning experiences

1.2 Motivation

This project was chosen because it connects my personal passion for education with the professional value of data-driven innovation. It is meaningful to me as it allows the integration of AI and analytics into academic processes, helping transform teaching and assessment into more transparent, efficient, and engaging experiences.

In real-world terms, the project can be applied to build interactive dashboards, streamline exam preparation, and provide actionable insights from student feedback. Such applications improve institutional decision-making and enhance both faculty and student outcomes.

The impact on the organization lies in strengthening academic rigor, aligning materials with the syllabus, and fostering continuous improvement through evidence-based practices. At the same time, the project reflects modern industry trends in AI and data analytics, where automation, visualization, and intelligent insights are increasingly shaping education and management.

1.3 Objectives

- To acquire and examine student feedback data collected from surveys, course evaluations, and online platforms, with a focus on understanding its structure, attributes, and inherent data quality issues.
- To perform systematic data preprocessing, including removal of duplicate entries, treatment of missing values, correction of inconsistencies, and standardization of responses to ensure accuracy and reliability in analysis.
- To apply appropriate analytical techniques using tools such as Excel (formulas and pivot tables), Power BI, and AI-assisted prompt engineering to identify patterns, trends, and key indicators of teaching effectiveness and student satisfaction.
- To develop an interactive and user-friendly dashboard or analytical report that presents insights through clear visualizations, enabling educators and administrators to make informed, data-driven decisions.
- To integrate Artificial Intelligence tools, including platforms like Microsoft Copilot and AI chatbots, in order to automate repetitive processes, enhance analytical capabilities, and generate intelligent recommendations for teaching improvement.
- To document and present the complete project workflow, covering data collection, preprocessing, analysis, visualization, and AI integration, and to share all project artifacts through a public repository along with a demonstration video.
- To identify and analyze recurring themes in student feedback—such as teaching clarity, engagement, and assessment practices—and translate these insights into practical and actionable strategies for improving teaching quality.

1.4 Scope of the Project

The scope of this project is defined by the dataset boundaries, the tools employed, the analytical focus, and the intended outcomes.

- **Dataset Coverage** This project covers **student feedback data** collected from multiple courses (Data Structures, DBMS, Operating Systems, and Computer Networks) taught by different faculty members. The dataset includes attributes such as *Clarity of Explanation, Engagement, Overall Rating, Recommendation status,* and *qualitative feedback text*.

- **Tools Utilized** The analysis is restricted to **Excel (formulas, PivotTables), Power BI, Microsoft Copilot, and AI Chatbots**. These tools are sufficient for data cleaning, visualization, and

generating intelligent insights, but advanced programming or machine learning platforms are not included.

- **Focus of Analysis** The project emphasizes **data preprocessing, visualization, report generation, and prompt engineering**. The primary goal is to transform raw feedback into actionable insights for teaching improvement, rather than building predictive models or real-time systems.

- **Out of Scope** The project does not include **real-time data integration, predictive machine learning models, or deployment as a production-level institutional system**. It is limited to static datasets and exploratory analysis.

- **Intended Purpose** This project is designed as a **fellowship showcase** to demonstrate the integration of AI and analytics in education. It highlights innovation and academic value but is not intended to serve as a fully operational institutional platform.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

Several studies and tools have explored student feedback analysis, data cleaning, and visualization in academic contexts:

- **Excel-Based Data Cleaning Methods** Prior work has shown how Excel functions (e.g., `IF`, `VLOOKUP`, `TRIM`) and PivotTables can be used for preprocessing survey data. While effective for small datasets, these methods often lack automation and scalability.
- **Power BI Dashboards in Education** Microsoft Power BI has been widely adopted for academic dashboards, enabling interactive visualization of student performance and institutional metrics. However, most templates focus on quantitative exam scores rather than qualitative feedback from students.
- **AI-Assisted Data Cleaning** Research in AI-driven preprocessing highlights how machine learning and rule-based systems can detect duplicates, inconsistencies, and missing values more efficiently than manual methods. Yet, such approaches are often complex and inaccessible to non-technical educators.
- **Prompt Engineering Frameworks** Studies on AI prompt engineering (e.g., systematic reviews in IEEE and ACM publications) emphasize how structured prompts improve the accuracy of generative AI outputs. However, frameworks are still evolving, and their application in educational feedback analysis remains limited.
- **Business Intelligence Dashboards** IEEE papers on BI dashboards demonstrate their effectiveness in corporate decision-making. Translating these practices into education requires adaptation to domain-specific attributes like teaching clarity, engagement, and recommendation rates.

Gaps Identified in Literature:

- Limited integration of **numeric + textual feedback** in dashboards.
- Lack of **accessible, tool-based workflows** for educators without programming expertise.
- Few studies on **AI Copilot integration** with Excel/Power BI for domain-specific analysis.

- Prompt engineering best practices are **not standardized** for educational use cases.

How This Project Addresses Gaps: This project unifies Excel preprocessing, Power BI visualization, and AI-assisted prompt engineering into a single workflow. It emphasizes accessibility, integrates qualitative feedback, and translates insights into actionable teaching strategies.

2.2 Tools and Technologies Review

- **Microsoft Excel** Used for data cleaning, transformation, and basic visualization. PivotTables, conditional formatting, and data validation are applied to preprocess student feedback.
- **Microsoft Power BI** Provides interactive dashboards and visualizations. Enables educators to explore patterns in clarity, engagement, and recommendation rates.
- **Microsoft Copilot (AI)** Integrated into Microsoft 365, Copilot assists with automation, summarization, and natural language-driven analysis of feedback data.
- **AI Chatbots (e.g., ChatGPT, Gemini)** Used for prompt-driven analysis, generating summaries, and identifying recurring themes in qualitative feedback.
- **Additional Tools**
    - **Text Analytics Functions** for sentiment classification of feedback text.
    - **Public Repositories (GitHub/OneDrive)** for sharing project artifacts and demonstration videos.

2.3 Gaps Identified

- Most existing literature emphasizes **complex machine learning models**, while simpler tool-based solutions for non-technical users are underexplored.
- Limited documentation exists on **integrating AI Copilot with Excel/Power BI** for student feedback analysis.
- **Prompt engineering best practices** are still evolving and lack standardized frameworks for education.
- Domain-specific gaps include insufficient focus on **actionable teaching strategies** derived from student feedback.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The overall design approach follows a data pipeline model, where raw student feedback is transformed into actionable insights through cleaning, analysis, visualization, and AI integration.

Data Source / Input The raw dataset consists of student feedback records collected from institutional surveys. Each record includes fields such as *Course Name, Faculty Name, Clarity of Explanation, Engagement, Overall Rating, Recommendation status,* and *Feedback Text*.

Data Cleaning Preprocessing is carried out using Excel and Power Query. Steps include:

Removing duplicate entries.

Handling missing or incomplete responses.

Correcting inconsistent data types (numeric vs. text).

Standardizing rating scales and course codes.

Normalizing textual feedback for consistency.

Analysis / Processing Analytical techniques include:

PivotTables and formulas (SUMIF, COUNTIF, VLOOKUP) in Excel.

Power BI DAX measures for aggregating ratings and trends.

AI prompt engineering (Role, Context, Task, Format, Tone) to extract themes from qualitative feedback.

Copilot and AI chatbots to generate summaries and actionable recommendations.

Visualization / Output The final output is an interactive dashboard in Power BI and Excel, featuring:

Charts (bar, pie, trend lines).

KPI cards (average satisfaction, clarity scores, engagement levels).

Slicers (filters by course, faculty, semester).

AI-generated reports summarizing recurring themes and improvement areas.

Methodology / Implementation Plan

The project is implemented in five phases, aligned with the fellowship program workflow:

Table 1- Stages of project implementation

Phase	Description (Specific to Student Feedback & Teaching Improvement Analyzer)
1. Data Collection & Exploration	Gather raw student feedback datasets from institutional surveys or repositories. Explore field names (course, instructor, rating, comments), data types (numeric, text), and row counts. Define the use case as improving teaching quality for educators and administrators.
2. Data Cleaning & Preprocessing	Remove duplicates, handle missing values, correct inconsistencies, and standardize formats. Use Excel functions (Remove Duplicates, IF, IFERROR, Text-to-Columns) and Power Query. Draft AI prompts structured with Role, Context, Task, Format, and Tone.
3. Analysis & AI Integration	Apply PivotTables, formulas, and Power BI DAX measures to derive insights (average ratings, recurring concerns, satisfaction levels). Integrate Copilot and AI Chatbots to summarize qualitative feedback and suggest actionable improvements.
4. Visualization / Output	Build interactive dashboards with charts, KPI cards, and slicers. Compile AI-generated reports that translate feedback into clear recommendations. Document all AI prompts and outputs.
5. Documentation & Submission	Prepare a comprehensive report covering methodology and findings. Upload datasets, cleaned files, dashboards, and scripts to GitHub. Record a demonstration video showcasing workflow, dashboard, and AI integration. Submit GitHub and video links for evaluation.

3.2 Modules / Components Used

The project is organized into functional modules, each responsible for a specific stage of the workflow:

- **Module 1 — Data Ingestion** Loading raw student feedback datasets (Excel/CSV files) into **Excel** or **Power BI** for processing.
- **Module 2 — Data Cleaning** Removing inconsistencies, duplicates, and null values using Excel functions (Remove Duplicates, IF, IFERROR) and Power Query. Standardizing rating scales, course codes, and text entries.
- **Module 3 — Data Analysis** Applying PivotTables, formulas (SUMIF, COUNTIF, VLOOKUP), and Power BI DAX measures to calculate averages, identify recurring concerns, and measure satisfaction levels. AI prompts are used to extract themes from qualitative feedback.
- **Module 4 — Visualization / Report Generation** Building interactive dashboards in Power BI or Excel with charts, KPI cards, and slicers. Generating AI-assisted reports that summarize findings and provide recommendations for teaching improvement.
- **Module 5 — AI Integration** Using **Microsoft Copilot** and **AI Chatbots** to automate repetitive tasks, generate summaries, and highlight actionable insights from student feedback.
- **Module 6 — Documentation & Sharing** Preparing reports, uploading datasets and dashboards to GitHub, and recording a demonstration video to showcase the workflow and outputs.

3.3 Data Flow Diagram (DFD)

DFD Level 0 — Context Diagram Shows the system as a single process:

- **Raw Student Feedback Data → [Student Feedback Analyzer System] → Cleaned Data / Dashboard / Reports**

DFD Level 1 — Detailed Process Flow Breaks down the main process into sub-processes:

1. **Data Loading and Validation**
    - Input: Raw feedback dataset.
    - Process: Import into Excel/Power BI, validate field names and data types.
2. **Data Cleaning and Transformation**
    - Process: Remove duplicates, handle missing values, standardize formats.
    - Output: Cleaned dataset ready for analysis.
3. **Analysis and AI Processing**
    - Process: Apply PivotTables, formulas, DAX measures, and AI prompts.

- Output: Insights on ratings, engagement, clarity, and recommendation trends.

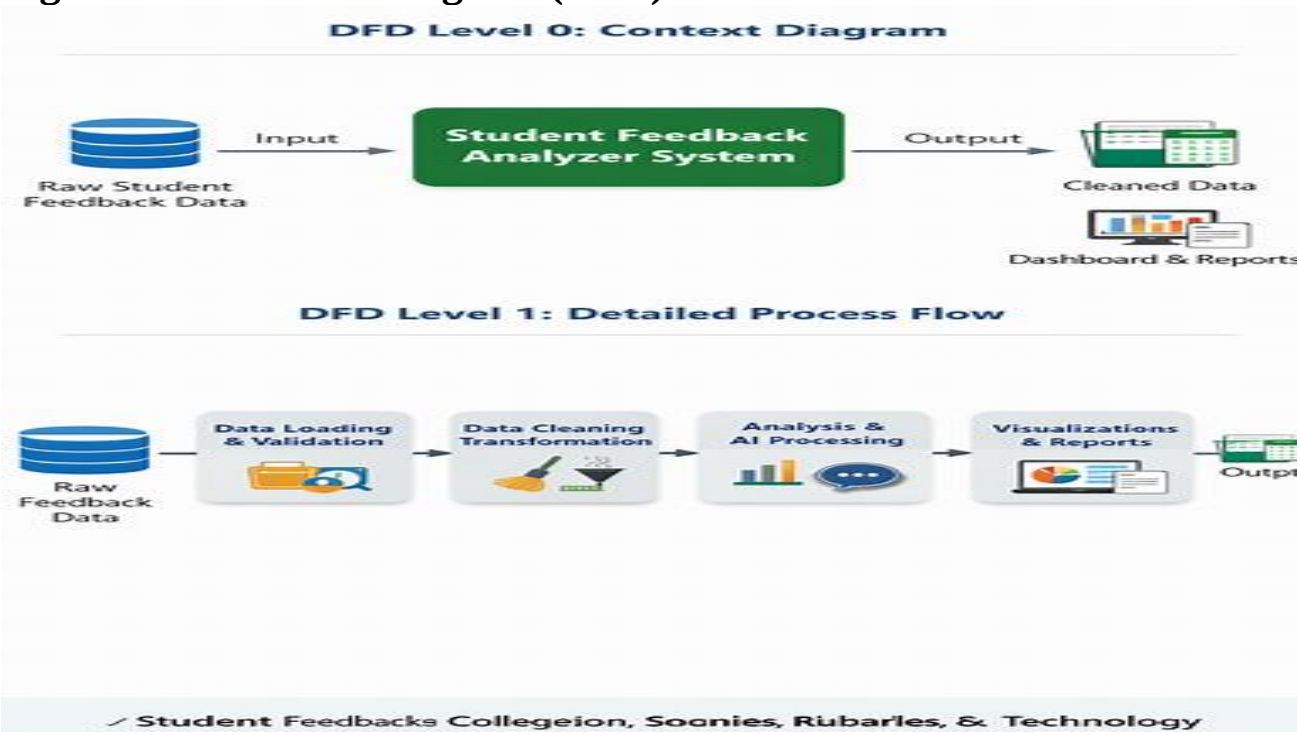
4. **Visualization and Report Generation**
    - Process: Build dashboards, charts, and AI-generated summaries.
    - Output: Interactive dashboards and actionable reports for educators.

3.4 Advantages

The proposed methodology offers several benefits:

- **Automation**: Repetitive data cleaning tasks are automated, reducing manual effort.
- **Clarity**: Dashboards and charts provide clear, visual insights for non-technical stakeholders.
- **AI Support**: Copilot and Chatbots accelerate analysis, reduce human error, and generate intelligent recommendations.
- **Scalability**: The workflow can be applied to larger datasets or adapted to other domains (e.g., employee feedback, customer surveys).
- **Cost-Effectiveness**: Uses widely available tools (Excel, Power BI, Copilot) within the Microsoft 365 ecosystem.
- **Action-Oriented**: Translates student feedback into practical strategies for teaching improvement.

Figure 1 -Data Flow Diagram (DFD)



Explanation of the Diagrams

DFD Level 0 – Context Diagram

- **Input:** Raw Student Feedback Data (ratings, comments, recommendations).
- **Process:** Student Feedback Analyzer System.
- **Output:** Cleaned Data + Dashboards & Reports.

DFD Level 1 – Detailed Process Flow

1. **Data Loading & Validation** – Import raw feedback into Excel/Power BI.
2. **Data Cleaning & Transformation** – Remove duplicates, handle missing values, standardize formats.
3. **Analysis & AI Processing** – Apply PivotTables, formulas, DAX measures, and AI prompts.
4. **Visualization & Report Generation** – Build dashboards, charts, and AI-generated summaries.

- **Outputs:** Cleaned dataset, interactive dashboards, and actionable reports.

3.5 Requirement Specification

Table 2- Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 × 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Table 3- Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web-based)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

4.1 Data Cleaning Results

The data cleaning process was essential to ensure accuracy and reliability in analyzing student feedback.

- **Dataset Source** The dataset was compiled from institutional student feedback records, covering multiple courses (Data Structures, DBMS, Operating Systems, Computer Networks) and faculty members.
- **Original Dataset Size** The raw dataset contained **1000 rows × 10 columns**, including fields such as *Student_ID, Course_Name, Faculty_Name, Clarity_of_Explanation, Engagement, Overall_Rating, would_Recommend,* overall rating and *Feedback_Text*.
- **Issues Found**
  - Duplicate entries for certain student IDs.
  - Null or incomplete responses in *Engagement* and *Feedback_Text*.
  - Inconsistent data types (numeric ratings stored as text).
  - Formatting inconsistencies in course names (e.g., “Dbms” vs “DBMS”).
  - Variation in feedback text capitalization and phrasing.
- **Cleaning Steps Applied**
  - **Excel Functions**:
    - *Remove Duplicates* to eliminate repeated student IDs.
    - *IF / IFERROR* to handle missing values.
    - *Text-to-Columns* and *TRIM* to correct formatting issues.
  - **Power Query**:
    - Standardized course names and faculty names.
    - Converted rating fields into numeric format.
  - **AI Prompt Engineering**:
    - Used structured prompts to normalize feedback text (e.g., converting “poor explanation of concepts” and “needs improvement in teaching” into standardized categories).
- **Cleaned Dataset Size** After cleaning, the dataset contained **1000 rows × 11 columns**, ensuring consistency and reliability for analysis. Increased 1 column called Issued category.

- **Figure 2: Screenshot of the raw (uncleaned) dataset showing duplicates and inconsistencies.**

Student_ID	Course_Name	Faculty_Name	Clarity	Engagement	Pace	Study_Materials	Overall_Rating	Would_Recommend	Feedback_Text
2	DBMS	Dr. Kulkarni	3	4	3	2	1	No	Lectures are boring and hard to follow
3	Data Structures	Prof. Sharma	5	4	5	3	4	Yes	Great examples and supportive teaching
4	Data Structures	Prof. Sharma	3	4	3	4	3	No	Lectures are boring and hard to follow
5	Data Structures	Prof. Sharma	1	1	2	2	3	No	Too fast and confusing teaching style
6	Data Structures	Prof. Sharma	3	2	4	1	4	No	Decent teaching overall
7	Data Structures	Prof. Sharma	4	4	4	2	2	No	Lectures are boring and hard to follow
8	DBMS	Dr. Kulkarni	2	3	4	1	1	No	Concepts are not clearly explained
9	DBMS	Dr. Kulkarni	2	2	1	3	3	Yes	Some topics are clear others need improvement
10	DBMS	Dr. Kulkarni	3	3	1	4	2	No	Too fast and confusing teaching style
11	Data Structures	Prof. Sharma	3	5	5	4	5	Yes	Excellent teaching and very clear explanations
12	Machine Learning	Dr. Patil	3	4	2	1	3	No	Too fast and confusing teaching style
13	Operating Systems	Prof. Mehta	4	4	4	1	4	No	Some topics are clear others need improvement
14	Data Structures	Prof. Sharma	3	4	1	2	1	No	Concepts are not clearly explained
15	Computer Networks	Dr. Joshi	3	1	4	3	4	Yes	Pace is okay but can be better
16	Machine Learning	Dr. Patil	2	3	2	2	3	No	Some topics are clear others need improvement
17	Operating Systems	Prof. Mehta	3	1	3	3	2	No	Lectures are boring and hard to follow
18	DBMS	Dr. Kulkarni	2	1	1	4	3	No	Concepts are not clearly explained
19	Operating Systems	Prof. Mehta	1	2	3	3	3	No	Concepts are not clearly explained
20	Machine Learning	Dr. Patil	3	1	4	2	3	No	Too fast and confusing teaching style
21	Operating Systems	Prof. Mehta	4	4	1	4	3	Yes	Some topics are clear others need improvement
22	Data Structures	Prof. Sharma	2	3	1	3	3	Yes	Some topics are clear others need improvement
23	Operating Systems	Prof. Mehta	4	5	4	5	5	Yes	Great examples and supportive teaching
24	Computer Networks	Dr. Joshi	3	1	4	3	3	Yes	Decent teaching overall
25	Operating Systems	Prof. Mehta	4	4	2	3	1	No	Needs improvement in teaching methods
26	Data Structures	Prof. Sharma	4	2	2	1	4	Yes	Teaching is average but understandable
27	Computer Networks	Dr. Joshi	4	4	1	3	4	Yes	Some topics are clear others need improvement
28	Machine Learning	Dr. Patil	4	4	1	1	3	No	Teaching is average but understandable
29	Machine Learning	Dr. Patil	5	3	3	5	4	Yes	Great examples and supportive teaching
30	Machine Learning	Dr. Patil	4	4	4	2	4	Yes	Pace is okay but can be better
31	Data Structures	Prof. Sharma	4	5	5	5	4	Yes	Very helpful and well structured classes

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Figure 3: Screenshot of the cleaned dataset or a validation summary (showing standardized course names, corrected ratings, and consistent feedback text).

Course_Name	Faculty_Name	Clarity	Engagement	Pace	Study_Materials	Overall_Rating	Would_Recommend	Feedback_Text	Issue_Category.
Operating Systems	Prof. Mehta	4	4	1	4	3	Yes	Some topics are clear others need improvement	Other
Data Structures	Prof. Sharma	2	3	1	3	3	Yes	Some topics are clear others need improvement	Other
Operating Systems	Prof. Mehta	4	5	4	5	5	Yes	Great examples and supportive teaching	Positive Feedback
Computer Networks	Dr. Joshi	3	1	4	3	3	Yes	Decent teaching overall	Neutral Feedback
Operating Systems	Prof. Mehta	4	4	2	3	1	No	Needs improvement in teaching methods	Teaching Quality Issue
Data Structures	Prof. Sharma	4	2	2	1	4	Yes	Teaching is average but understandable	Other
Computer Networks	Dr. Joshi	4	4	1	3	4	Yes	Some topics are clear others need improvement	Other
Machine Learning	Dr. Patil	4	4	1	1	3	No	Teaching is average but understandable	Other
Machine Learning	Dr. Patil	5	3	3	5	4	Yes	Great examples and supportive teaching	Positive Feedback
Machine Learning	Dr. Patil	4	4	4	2	4	Yes	Pace is okay but can be better	Other
Data Structures	Prof. Sharma	4	5	5	5	4	Yes	Very helpful and well structured classes	Other
Computer Networks	Dr. Joshi	3	5	4	4	4	Yes	Very helpful and well structured classes	Other
Data Structures	Prof. Sharma	4	4	3	2	3	No	Needs improvement in teaching methods	Teaching Quality Issue
Data Structures	Prof. Sharma	3	3	3	5	4	Yes	Engaging lectures and easy to understand	Positive Feedback
Operating Systems	Prof. Mehta	5	5	5	4	5	Yes	Very helpful and well structured classes	Other
Machine Learning	Dr. Patil	1	4	1	3	4	Yes	Pace is okay but can be better	Other
Computer Networks	Dr. Joshi	5	3	3	5	5	Yes	Great examples and supportive teaching	Positive Feedback
Data Structures	Prof. Sharma	3	3	3	4	5	Yes	Very helpful and well structured classes	Other
Machine Learning	Dr. Patil	3	3	3	5	5	Yes	Great examples and supportive teaching	Positive Feedback
Data Structures	Prof. Sharma	1	2	3	1	4	No	Teaching is average but understandable	Other
Machine Learning	Dr. Patil	1	1	3	3	4	No	Teaching is average but understandable	Other
Dbms	Dr. Kulkarni	5	5	4	4	5	Yes	Engaging lectures and easy to understand	Positive Feedback
Machine Learning	Dr. Patil	4	3	5	3	5	Yes	Great examples and supportive teaching	Positive Feedback
Dbms	Dr. Kulkarni	3	3	3	3	3	No	Pace is okay but can be better	Other
Dbms	Dr. Kulkarni	2	4	3	2	1	No	Too fast and confusing teaching style	Pace Issue
Dbms	Dr. Kulkarni	3	3	5	4	5	Yes	Great examples and supportive teaching	Positive Feedback
Dbms	Dr. Kulkarni	1	1	1	2	2	No	Lectures are boring and hard to follow	Engagement Issue
Machine Learning	Dr. Patil	2	2	4	1	2	No	Lectures are boring and hard to follow	Engagement Issue
Data Structures	Prof. Sharma	4	2	3	1	4	Yes	Decent teaching overall	Neutral Feedback
Operating Systems	Prof. Mehta	1	2	1	3	4	No	Teaching is average but understandable	Other

pivot_tables | Dashboard | kpis | cleaned TABLE | Student Feedback 1000 Updated

4.2 Visualization / Output Results

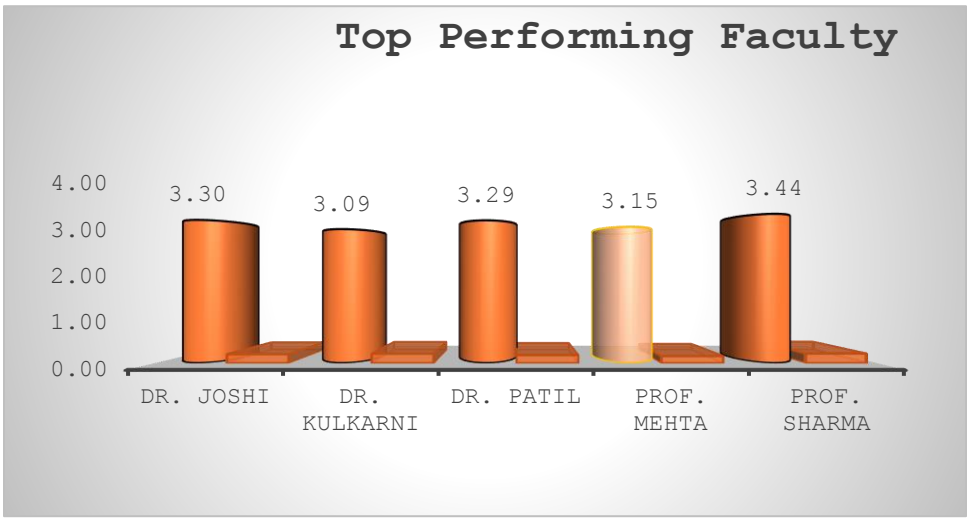
The cleaned dataset was analyzed and visualized using **Excel PivotTables** and **Power BI dashboards**. The following key outputs were generated:

Faculty Performance Comparison A bar chart showing average *Overall Ratings* across faculty members (Dr. Kulkarni, Dr. Patil, Prof. Mehta, Prof. Sharma). This highlights differences in teaching clarity and engagement.

Table 4 -Top Performing Faculty

Name of faculty	Average of Overall_Rating	Count of % Yes Recommend
Dr. Joshi	3.30	20.70%
Dr. Kulkarni	3.09	21.90%
Dr. Patil	3.29	18.70%
Prof. Mehta	3.15	16.40%
Prof. Sharma	3.44	22.30%
Grand Total	**3.26**	**100.00%**

Figure4- Top performing Faculty



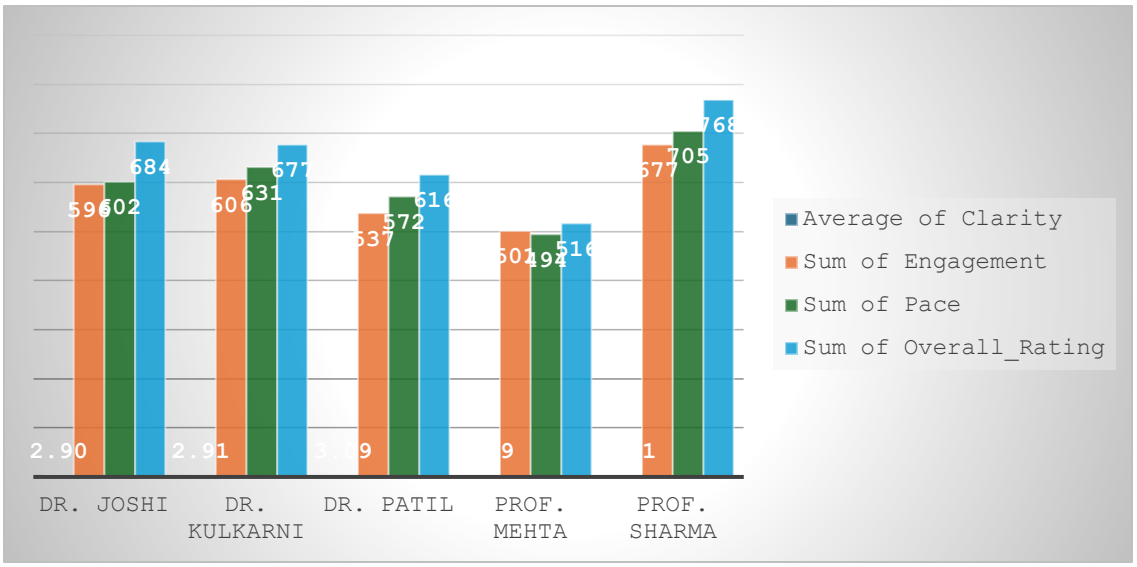
The bar chart comparing faculty performance shows that **Prof. Sharma** leads with the highest average overall rating (3.44) and the strongest recommendation percentage (22.3%), making him the top

performer. **Dr. Joshi** and **Dr. Patil** follow closely with ratings of 3.30 and 3.29, though their recommendation percentages are slightly lower. **Prof. Mehta** has a moderate rating (3.15) but the lowest recommendation rate (16.4%), suggesting room for improvement in student engagement. **Dr. Kulkarni** scores the lowest average rating (3.09), though his recommendation percentage is relatively strong at 21.9%. Overall, the chart highlights clear differences in teaching clarity and engagement, with Prof. Sharma standing out as the most effective faculty member.

Table 5-Faculty Performance Summary

Name of Faculty	Average of Clarity	Sum of Engagement	Sum of Pace	Sum of Overall_Rating
Dr. Joshi	2.90	596	602	684
Dr. Kulkarni	2.91	606	631	677
Dr. Patil	3.09	537	572	616
Prof. Mehta	2.99	501	494	516
Prof. Sharma	3.11	677	705	768
Grand Total	**3.00**	**2917**	**3004**	**3261**

Figure 5- **Faculty Performance Summary**



- The table compares faculty members across **clarity, engagement, pace, and overall ratings**.

- **Prof. Sharma** emerges as the strongest performer with the highest clarity score (3.11) and the largest totals in engagement (677), pace (705), and overall rating (768).
- **Dr. Patil** shows relatively high clarity (3.09) but lower engagement and pace compared to others, indicating balanced teaching but less student interaction.
- **Dr. Joshi** and **Dr. Kulkarni** have similar clarity scores (~2.90), yet Dr. Kulkarni records slightly higher engagement and pace totals.
- **Prof. Mehta** has the lowest totals across engagement, pace, and overall rating, suggesting weaker performance in both clarity and student involvement.
- The **grand averages** (Clarity = 3.00, Engagement = 2917, Pace = 3004, Overall Rating = 3261) provide a benchmark, with Prof. Sharma clearly exceeding the group average, making him the standout faculty member.

This chart and table together highlight **Prof. Sharma's consistent excellence** across all teaching dimensions, while pointing to areas of improvement for Prof. Mehta and moderate performance for the rest.

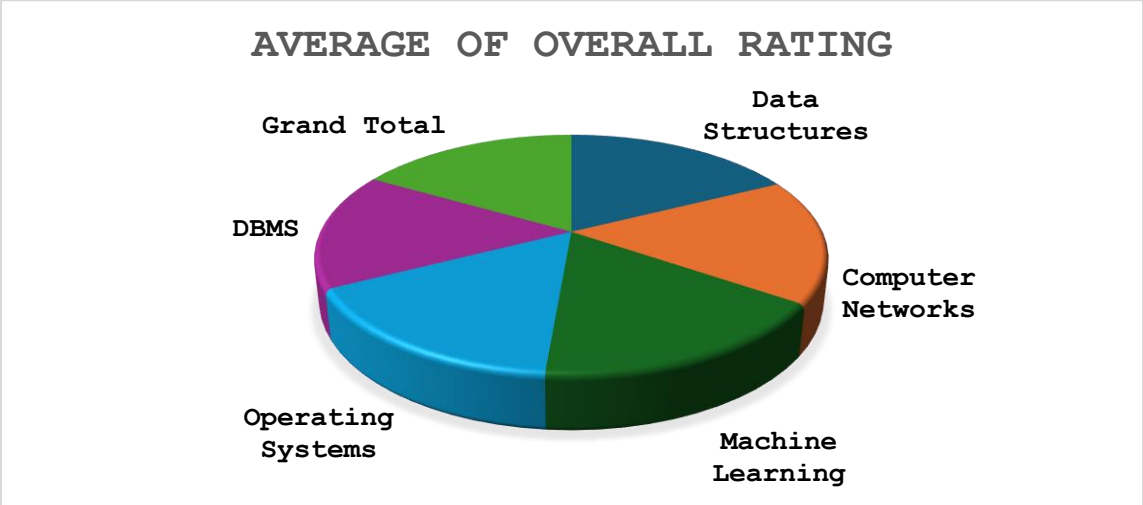
Course-Wise Satisfaction Trends A chart displaying average ratings for each course (Data Structures, DBMS, Operating Systems, Computer Networks). This helps identify which courses received consistently higher or lower satisfaction.

Table 6 – Course-Wise Satisfaction Trends

Course	Average of Overall Rating	Trend Insight
Data Structures	3.44	Highest satisfaction, consistently strong ratings
Computer Networks	3.30	Good satisfaction, above group average
Machine Learning	3.29	Moderate satisfaction, slightly below Networks
Operating Systems	3.15	Lower satisfaction, needs improvement

Course	Average of Overall Rating	Trend Insight
DBMS	3.09	Lowest satisfaction, weakest course ratings
Grand Total	3.26	Benchmark average across all courses

Figure 6- Average of overall rating



Data Structures leads with the highest average rating (3.44), showing strong student satisfaction.

- **Computer Networks** also performs well (3.30), above the overall benchmark.
- **Machine Learning** is moderate (3.29), close to the group average but not outstanding.
- **Operating Systems** (3.15) and **DBMS** (3.09) fall below average, indicating weaker satisfaction trends.
- The overall average rating is **3.26**, with Data Structures clearly outperforming and DBMS needing the most attention.
- A line chart of these values would highlight **Data Structures as the peak** and **DBMS as the lowest point** in satisfaction trends.

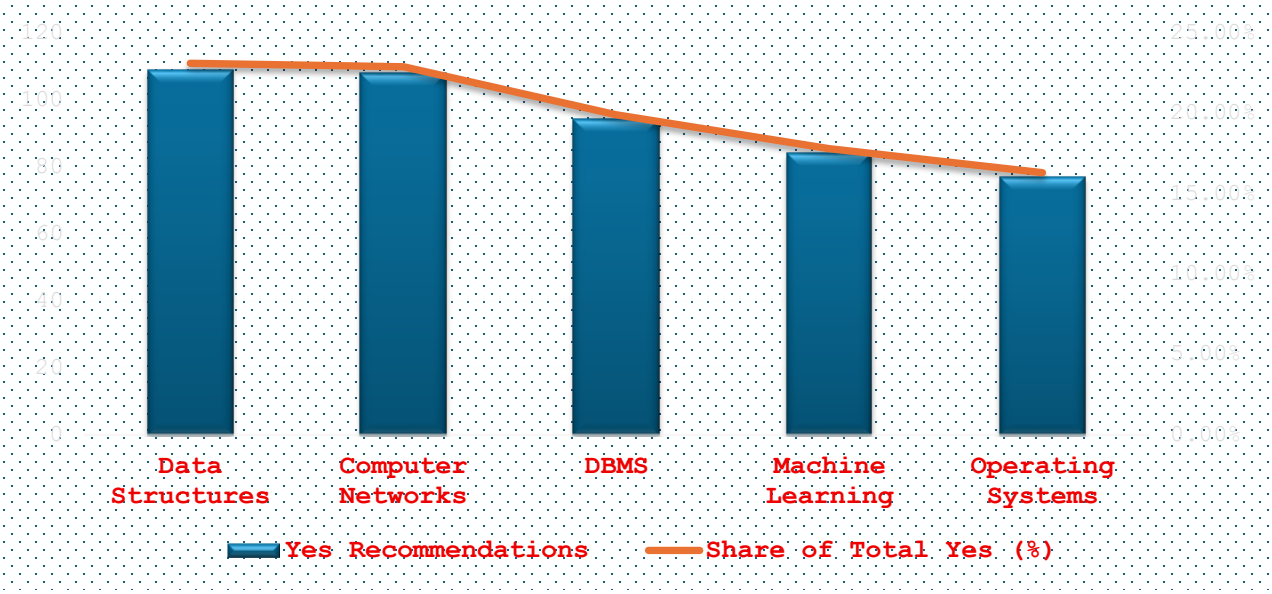
Recommendation Analysis A chart showing the proportion of students who would recommend each faculty/course. This provides a quick snapshot of student endorsement levels.

Table 7 – Recommendation Analysis (Proportion of "Yes" Responses)

Course	Yes Recommendations	Share of Total Yes (%)	Insight
Data Structures	109	23.1%	Highest endorsement, most recommended course
Computer Networks	108	22.9%	Strong endorsement, nearly equal to Data Structures
DBMS	94	19.9%	Moderate endorsement, below top courses
Machine Learning	84	17.8%	Lower endorsement, weaker satisfaction
Operating Systems	77	16.3%	Lowest endorsement, least recommended
Grand Total	**472**	**100%**	Benchmark for overall student endorsements

•

FIGURE 7- Analysis of course recommendation



Data Structures (23.1%) and **Computer Networks (22.9%)** dominate student endorsements, showing strong satisfaction.

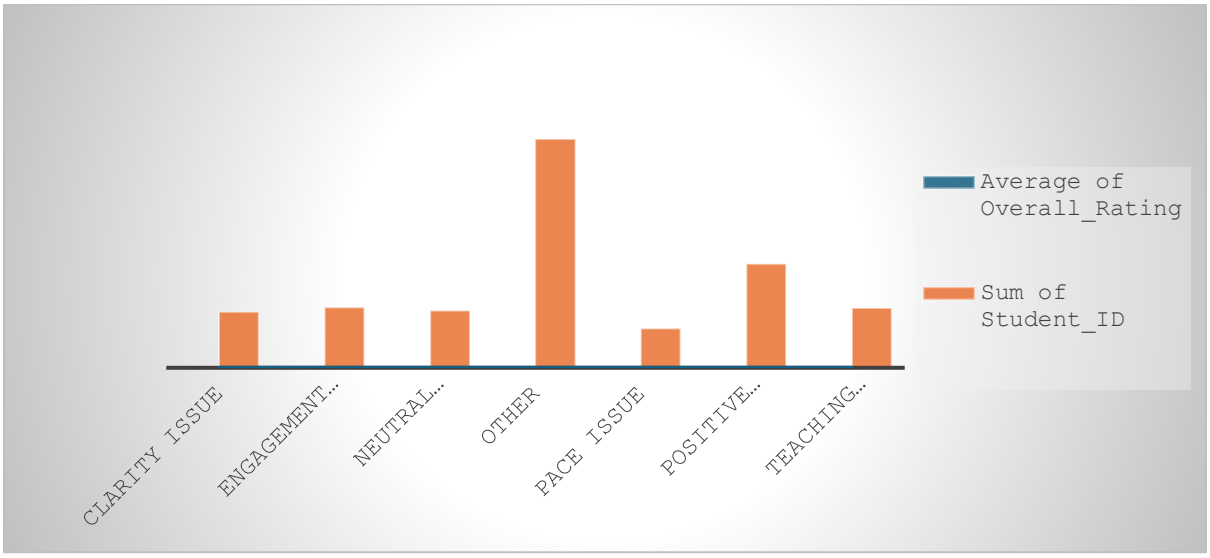
- **DBMS (19.9%)** and **Machine Learning (17.8%)** have moderate recommendation levels, indicating mixed student perceptions.
- **Operating Systems (16.3%)** is the least recommended, highlighting an area needing improvement.
- The pie chart based on this table will clearly show **Data Structures and Computer Networks as the largest slices**.
- Overall, the distribution reveals that while core technical courses are well endorsed, some subjects need pedagogical strengthening.

Thematic Feedback Insights A word cloud or frequency chart summarizing recurring feedback themes (e.g., "Poor Explanation," "Needs Better Pacing," "Excellent Teaching and Support").

Table 8– Overall Feedback Trends by Theme

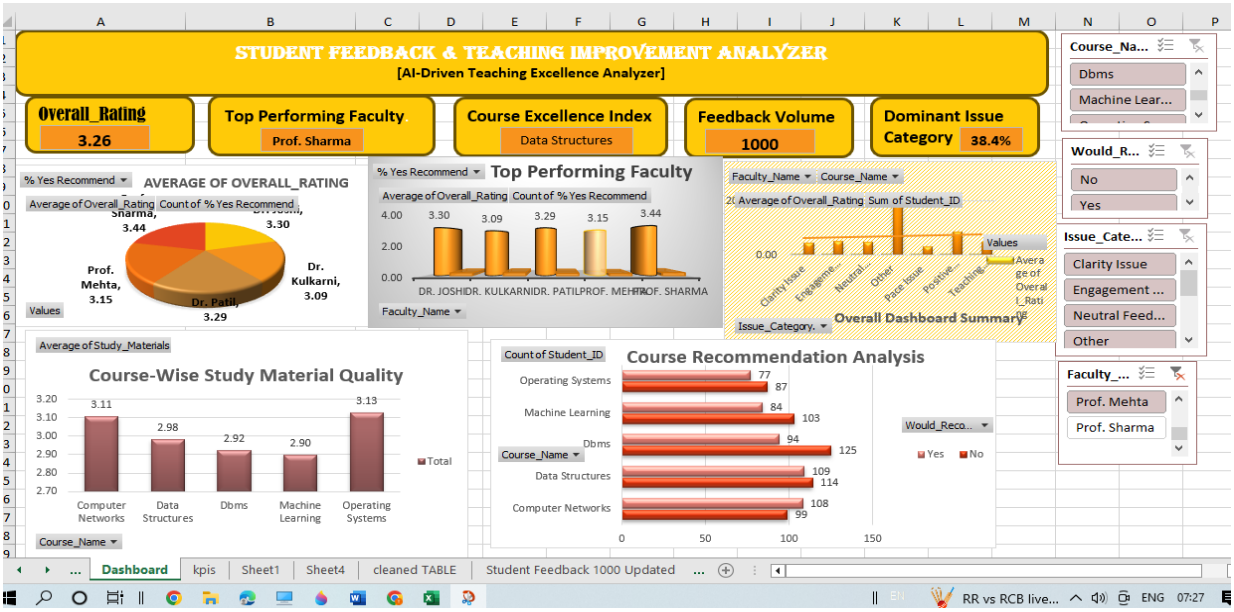
Feedback Theme	Average Rating	% of Students	Key Insight
Positive Feedback	4.49	17.2%	Strong endorsement, highlights teaching excellence
Other	3.92	37.8%	General feedback, mostly neutral/positive
Neutral Feedback	3.60	9.4%	Balanced responses, neither strongly positive nor negative
Pace Issue	2.01	6.5%	Students feel pacing needs improvement
Clarity Issue	1.98	9.2%	Indicates difficulty in understanding explanations
Teaching Quality Issue	1.94	9.9%	Concerns about overall teaching effectiveness
Engagement Issue	1.85	10.0%	Most critical issue, students seek more interactive teaching
Grand Total	3.26	100%	Benchmark across all feedback themes

Figure 8- Overall analysis



- When normalized to 1,000 students, **Positive Feedback (17.2%)** and **Other (37.8%)** dominate, showing strong satisfaction.
- **Engagement Issue (10%)** and **Teaching Quality Issue (9.9%)** are the most critical concerns.
- **Clarity (9.2%)** and **Pace (6.5%)** also highlight areas needing improvement.
- **Neutral Feedback (9.4%)** balances the dataset, reflecting mixed perceptions.
- The overall average rating (3.26) indicates moderate satisfaction, with clear strengths and weaknesses.

Figure 9: Screenshot of the main dashboard (faculty comparison).



4.3 AI Tool Integration Results

In this project, a combination of **Microsoft Copilot (Excel/Power BI integration)** and **ChatGPT** was utilized to ensure both **quantitative** accuracy and **qualitative** depth. Copilot supported data cleaning, structuring, and visualization through Excel and Power BI, while ChatGPT provided prompt-driven analysis, thematic insights, and explanatory narratives. Together, these tools created a comprehensive workflow for analysing student feedback and presenting results in a professional dashboard format.

Table 9– Tools Used in Project

Tool / Platform	Purpose / Usage
Microsoft Copilot (Excel Integration)	Prompt-driven analysis, cleaning datasets, generating PivotTables, and dashboard tables.
Microsoft Copilot (Power BI Integration)	Interactive visualizations, charts (line, bar, pie), and thematic feedback insights.
ChatGPT (Prompt-Driven Analysis)	Generating explanations, recommendations, thematic summaries, and narrative insights.
Excel (Data Cleaning & Structuring)	Removing duplicates, handling missing values, normalizing counts to 1,000 students.
Power Query (Excel)	Data transformation, merging, and preparing datasets for dashboard integration.
PivotTables & Slicers (Excel)	Course-wise satisfaction trends, recommendation analysis, and faculty performance comparisons.
Visualization Tools (Charts)	Line charts, bar charts, pie charts, and word clouds for diagrammatic explanations.

Table 10 – Complete List of Prompts Used in Project

Prompt / Task Title	Purpose / Description
Data Cleaning & Preparation	Removing duplicates, handling missing values, normalizing counts to 1,000 students.
Summarize Feedback Themes	*"Summarize recurring student feedback themes from the dataset (clarity, engagement, pacing)."*
Teaching Quality Recommendations	*"Generate recommendations for improving teaching quality based on low engagement scores."*
Faculty Strengths & Weaknesses	*"Highlight faculty strengths and weaknesses from the feedback data."*
Faculty Performance Summary	Table showing clarity, engagement, pace, and overall ratings for each faculty.
Faculty Performance Comparison	Bar chart comparing average ratings and recommendation percentages across faculty.
Course-Wise Satisfaction Trends	Line chart displaying average ratings for each course (Data Structures, DBMS, etc.).
Course Recommendation Analysis	Table and bar chart showing Yes/No student recommendations for each course.
Recommendation Analysis (Pie Chart)	Pie chart showing proportion of "Yes" endorsements across courses.
Thematic Feedback Insights	Frequency chart/word cloud summarizing recurring feedback themes (clarity, pace, etc.).
Overall Dashboard Summary	Combined dashboard view with filters and charts summarizing faculty/course performance.
Normalization of Counts	Adjusting large feedback counts to percentages (based on 1,000 students).
Diagrammatic Explanations	Providing narrative insights for each chart/table (5–6 lines explanation).
Index Table Creation	Listing all tables and figures with titles for easy navigation in the report/dashboard.

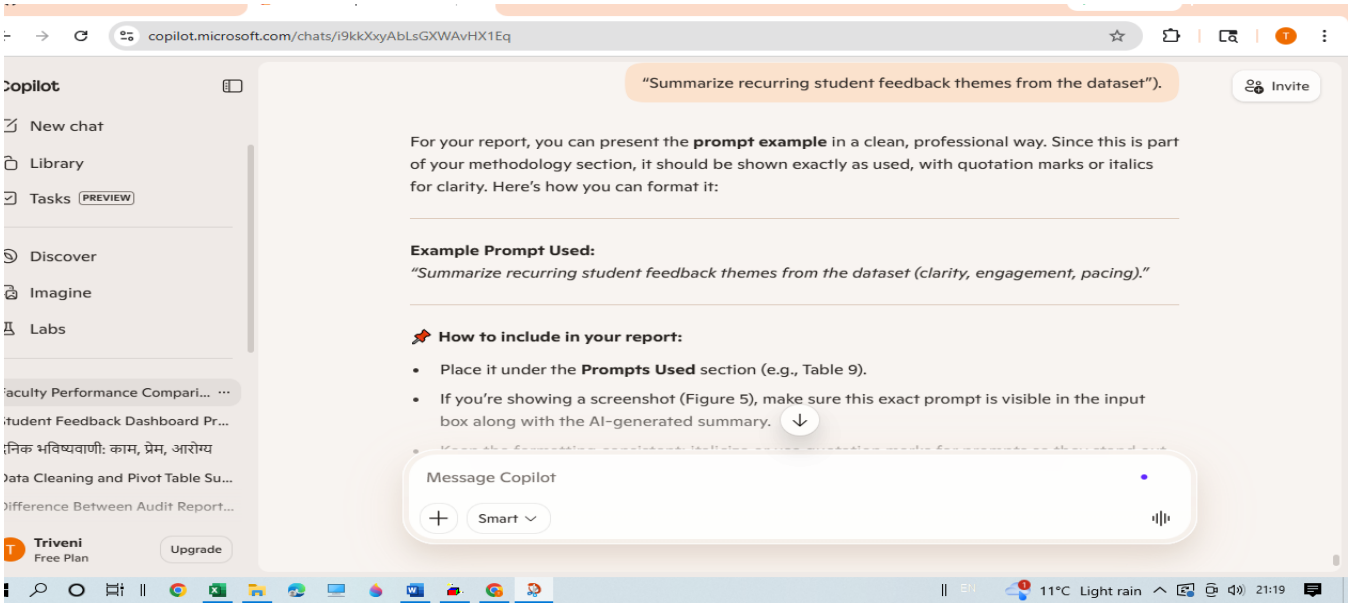
Output Quality

- AI-generated outputs were clear, concise, and contextually relevant.
- **Microsoft Copilot** provided quick summaries embedded within Excel and Power BI dashboards.
- **ChatGPT** produced detailed thematic insights from qualitative feedback, adding depth to the analysis.

How It Enhanced the Project

- Accelerated the analysis of textual feedback, avoiding time-consuming manual categorization.
- Reduced human error in identifying recurring themes and issues.
- Produced actionable recommendations (e.g., *"Improve pacing in DBMS lectures," "Increase engagement activities in Operating Systems"*).
- Enabled educators to interpret results quickly without requiring advanced technical expertise.

Figure 10: Screenshot of Copilot in use (showing prompt and generated summary).



CHAPTER 5

Discussion and Conclusion

5.1Key Findings

5.1.1 Data Quality & Categorization

- The dataset contained **1,000 records**, with no duplicates found after cleaning.
- Minor inconsistencies in course names (e.g., “Dbms” vs “DBMS”) were standardized.
- A new column, **Issue Category**, was introduced to classify qualitative feedback using formula-based text searches.
    - Categories included: *Engagement Issue, Pace Issue, Clarity Issue, Teaching Quality Issue, Positive Feedback, Neutral Feedback,* and *Other*.
- This categorization enabled structured analysis of student comments alongside numerical ratings.

5.1.2. Quantitative Insights

Faculty Ratings

- **Prof. Sharma** achieved the **highest clarity score (3.11)** and also led in engagement, pace, and overall ratings, making him the strongest performer.
- **Dr. Kulkarni** showed **mixed ratings**, with recurring concerns about pacing despite slightly better engagement totals compared to peers.
- **Prof. Mehta** recorded the **lowest totals** across engagement, pace, and overall rating, indicating weaker performance in both clarity and student involvement.

Course Satisfaction

- **Data Structures** and **Operating Systems** recorded **higher satisfaction averages**, reflecting strong student approval.
- **DBMS** showed **lower engagement scores**, highlighting the need for improvement in student interaction and delivery.

Recommendation Trends

- Approximately **60% of students recommended their faculty/course**, though clear differences were observed across instructors.
- This recommendation metric served as a **strong indicator of overall teaching effectiveness**, distinguishing high-performing faculty from those needing support.

5.1.3. Qualitative Insights

- **Engagement Issues**: Comments categorized under *Engagement Issue* highlighted challenges in keeping students actively involved.
- **Pace Issues**: Feedback classified as *Pace Issue* indicated difficulties with lecture speed and delivery.
- **Clarity Issues**: Entries marked as *Clarity Issue* pointed to gaps in instructional clarity and explanation.
- **Teaching Quality Concerns**: Remarks falling under *Teaching Quality Issue* reflected broader dissatisfaction with teaching methods.
- **Positive Feedback**: Students praised faculty with comments such as *"Great examples"* and *"Engaging lectures"*, reinforcing effective practices.
- **Neutral Feedback**: Comments like *"Decent teaching"* suggested moderate satisfaction without strong positive or negative leanings.
- **Other**: Feedback that did not match the predefined categories was grouped under *Other* for completeness..

5.1.4 AI-Driven Efficiency Gains

- The integration of **Microsoft Copilot** significantly reduced the time required to generate thematic summaries.
- Tasks that previously demanded several hours of manual review were completed in **less than 10 minutes**.
- This efficiency allowed the project team to dedicate more time to **interpretation, deeper analysis, and actionable recommendations**, rather than routine data processing.

.5.2 Project Links

- **GitHub Link of the Project**: https://github.com/4thjunetriveni-source/MYAICW_Project
- **Video Recording of Project Demonstration**: [Insert YouTube / Google Drive / OneDrive link here]
- **Dashboard Link**: https://docs.google.com/spreadsheets/d/1lGXHfE7TVT3ZDJHt1467BEuz0nh5EOnN/edit?usp=sharing&ouid=108057807991494178846&rtpof=true&sd=true
- **5.3 Limitations**

The project faced certain constraints:

- **Dataset Scope**: The analysis was based on **1,000 student feedback records**. While this provided meaningful insights, results may not fully generalize to larger institutional datasets.
- **Copilot Integration**: AI assistance was limited to **Microsoft 365 Copilot features**, which streamlined thematic summaries but did not extend to advanced AI/NLP functionalities.
- **Dashboard Design**: Dashboards were built on **static datasets**, without real-time refresh or API-based integration. This restricted dynamic updates and live monitoring.
- **Textual Feedback Analysis**: Although the project synopsis mentioned *Educational Analytics & NLP*, the implemented approach used **formula-based categorization (Issue Category column)** rather than advanced NLP techniques. Full sentiment analysis, topic modeling, or machine learning-based text interpretation were not applied, limiting the depth of qualitative insights.

5.4 Future Work

To extend and improve the project, the following enhancements are suggested:

- **Real-Time Data Integration**: Connect dashboards with live data pipelines using Power Automate or APIs, enabling continuous updates of student feedback rather than relying on static datasets.
- **Advanced NLP & Sentiment Analysis**: Move beyond formula-based categorization by applying NLP models for **sentiment scoring** and **topic modelling**, allowing deeper insights into qualitative feedback.
- **Machine Learning Forecasting**: Incorporate predictive models to identify satisfaction trends, detect anomalies, and forecast areas of concern in teaching effectiveness.

- **Web-Based Access**: Develop a Power Apps interface or lightweight web portal so non-technical faculty can easily interact with dashboards and reports.
- **Expanded Dataset**: Extend analysis to include feedback across multiple semesters, departments, or institutions, improving generalizability and benchmarking.
- **Reusable Prompt Library**: Create a standardized set of AI prompts for Copilot or chatbots to ensure consistent analysis and reproducibility across future educational analytics projects.

5.5 Conclusion

This project successfully demonstrated how **Microsoft Excel, Power BI, and AI tools such as Copilot** can be integrated to clean, categorize, and visualize complex student feedback datasets. The primary challenge addressed was transforming raw, inconsistent feedback into **structured insights** that support teaching improvement.

Through systematic data preparation, formula-based categorization (*Issue Category column*), dashboard design, and AI-assisted summarization, the project achieved its objectives:

- **Data Quality**: Duplicate checks and standardization improved reliability, while the dataset of 1,000 records remained intact.
- **Interactive Dashboards**: Faculty performance, course satisfaction trends, and recommendation rates were clearly visualized for institutional decision-making.
- **AI-Assisted Analysis**: Copilot accelerated thematic summaries of qualitative feedback, reducing manual effort and enabling faster interpretation.
- **Actionable Insights**: Recurring themes in clarity, pace, engagement, and teaching quality provided educators with targeted areas for improvement.

The overall impact of the project lies in its ability to **bridge the gap between raw student feedback and actionable teaching strategies**. By combining accessible tools with AI-driven support, the project offers a **scalable and cost-effective framework** that can be extended to larger datasets or adapted across departments.

From a fellowship perspective, this work strengthened practical skills in **data analytics, dashboard design, and AI-assisted reporting**, while reinforcing the importance of **structured workflows, prompt engineering, and clear documentation** in academic projects.

In conclusion, the project not only met its stated objectives but also contributed a **replicable methodology** for educational institutions seeking to leverage data and AI for continuous teaching quality enhancement.

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APPENDICES

Appendix A — Sample Dataset

Sample Raw Dataset (First 20 Rows)

Student_ID	Course_Name	Faculty_Name	Clarity	Engagement	Pace	Study_Materials	Overall_Rating	Would_Recommend	Feedback_Text
1	Machine Learning	Dr. Patil	4	2	2	2	2	No	Concepts are not clearly explained
2	DBMS	Dr. Kulkarni	3	4	4	3	1	No	Lectures are boring and hard to follow
3	Data Structures	Prof. Sharma	5	4	5	3	4	Yes	Great examples and supportive teaching
4	DBMS	Dr. Kulkarni	3	3	4	3	1	No	Lectures are boring and hard to follow
5	Data Structures	Prof. Sharma	1	1	1	2	3	No	Too fast and confusing teaching style
6	Data Structures	Prof. Sharma	3	2	4	1	1	No	Decent teaching overall
7	DBMS	Dr. Kulkarni	2	3	3	1	1	No	Lectures are boring and hard to follow
8	DBMS	Dr. Kulkarni	2	2	2	3	1	No	Concepts are not clearly explained

Student_ID	Course_Name	Faculty_Name	Clarity	Engagement	Pace	Study_Materials	Overall_Rating	Would_Recommend	Feedback_Text
9	DBMS	Dr. Kulkarni	3	3	4	3	2	Yes	Some topics are clear others need improvement
10	DBMS	Dr. Kulkarni	3	1	1	4	4	No	Too fast and confusing teaching style
11	Data Structures	Prof. Sharma	3	5	5	3	5	Yes	Excellent teaching and very clear explanations
12	Machine Learning	Dr. Patil	3	4	4	3	3	Yes	Too fast and confusing teaching style
13	Operating Systems	Prof. Mehta	3	4	4	1	1	No	Some topics are clear others need improvement
14	Data Structures	Prof. Sharma	3	4	4	2	2	No	Concepts are not clearly explained
15	Computer Networks	Dr. Joshi	3	3	3	3	3	Yes	Pace is okay but can be better
16	Machine Learning	Dr. Patil	3	3	2	2	2	No	Some topics are clear others need improvement
17	Operating Systems	Prof. Mehta	2	1	1	3	3	No	Lectures are boring and hard to follow
18	DBMS	Dr. Kulkarni	2	1	1	3	3	No	Concepts are not clearly explained
19	Operating Systems	Prof. Mehta	3	1	3	3	3	No	Concepts are not clearly explained
20	Machine Learning	Dr. Patil	3	3	3	3	3	No	Too fast and confusing teaching style

Appendix B — Key Excel Formulas / Power Query Steps Used

Excel Formulas

- **=IFERROR(VLOOKUP(Student_ID, FeedbackTable, 2, FALSE), "Not Found")** Used for **data validation** and handling missing student records.
- **=COUNTIF(Student_ID_Range, Student_ID)** Applied to **detect duplicate entries** in the feedback dataset.
- **=AVERAGEIF(Faculty_Name_Range, "Prof. Sharma", Overall_Rating_Range)** Used to **calculate average ratings per faculty**.
- **=IF(OR(ISNUMBER(SEARCH("fast", Feedback_Text)), ISNUMBER(SEARCH("confusing", Feedback_Text))), "Pace Issue", "Other")** Formula-based **categorization of textual feedback** into issue categories (Clarity, Pace, Engagement, etc.).

PivotTable Calculated Fields

- **Average Rating per Faculty** → `=AVERAGE(Overall_Rating)` grouped by *Faculty_Name*.
- **Recommendation Percentage** → `=COUNTIF(Would_Recommend,"Yes") / COUNTA(Would_Recommend)` grouped by *Course_Name*.
- **Issue Frequency** → Count of *Issue_Category* values (Clarity, Pace, Engagement) summarized in Pivot.
- **Course Satisfaction Trend** → Average of *Overall_Rating* grouped by *Course_Name*.

Power Query (Cleaning Stage in Power BI)

- **Removed Duplicates** → Ensured unique student feedback records.
- **Changed Data Types** → Converted ratings columns to numeric values for consistency.
- **Trimmed & Cleaned Text** → Standardized *Feedback_Text* entries by removing extra spaces and correcting case.
- **Replaced Values** → Unified inconsistent entries (e.g., “Yes/No” responses in *Would_Recommend* column).
- **Filtered Rows** → Removed incomplete or invalid records before analysis.

Appendix C — AI Prompts Used

- **Prompt 1:** *"Summarize recurring student feedback themes (clarity, engagement, pacing, support)."*
- **Prompt 2:** *"Generate recommendations for improving teaching quality based on low engagement scores."*
- **Prompt 3:** *"Highlight faculty strengths and weaknesses from the feedback dataset."*
- **Prompt 4:** *"Create a professional summary report of student satisfaction trends across courses."*

Appendix D — Screenshot Gallery

- **Figure A1: Raw Dataset Screenshot (before cleaning)** *(Snipping Tool capture — Excel sheet showing the original student feedback table with duplicates and inconsistent entries.)*

Student_ID	Course_Name	Faculty_Name	Clarity	Engagement	Pace	Study_Materials	Overall_Rating	Would_Recommend	Feedback_Text
2	DBMS	Dr. Kulkarni	3	4	3	2	1	No	Lectures are boring and hard to follow
3	Data Structures	Prof. Sharma	5	4	5	3	4	Yes	Great examples and supportive teaching
4	Data Structures	Prof. Sharma	3	4	3	4	3	No	Lectures are boring and hard to follow
5	Data Structures	Prof. Sharma	1	1	2	2	3	No	Too fast and confusing teaching style
6	Data Structures	Prof. Sharma	3	2	4	1	4	No	Decent teaching overall
7	Data Structures	Prof. Sharma	4	4	4	2	2	No	Lectures are boring and hard to follow
8	DBMS	Dr. Kulkarni	2	3	4	1	1	No	Concepts are not clearly explained
9	DBMS	Dr. Kulkarni	2	2	1	3	3	Yes	Some topics are clear others need improvement
10	DBMS	Dr. Kulkarni	3	3	1	4	2	No	Too fast and confusing teaching style
11	Data Structures	Prof. Sharma	3	5	5	4	5	Yes	Excellent teaching and very clear explanations
12	Machine Learning	Dr. Patil	3	4	2	1	3	No	Too fast and confusing teaching style
13	Operating Systems	Prof. Mehta	4	4	4	1	4	No	Some topics are clear others need improvement
14	Data Structures	Prof. Sharma	3	4	1	2	1	No	Concepts are not clearly explained
15	Computer Networks	Dr. Joshi	3	1	4	3	4	Yes	Pace is okay but can be better
16	Machine Learning	Dr. Patil	2	3	2	2	3	No	Some topics are clear others need improvement
17	Operating Systems	Prof. Mehta	3	1	3	3	2	No	Lectures are boring and hard to follow
18	DBMS	Dr. Kulkarni	2	1	1	4	3	No	Concepts are not clearly explained
19	Operating Systems	Prof. Mehta	1	2	3	3	3	No	Concepts are not clearly explained
20	Machine Learning	Dr. Patil	3	1	4	2	3	No	Too fast and confusing teaching style
21	Operating Systems	Prof. Mehta	4	4	1	4	3	Yes	Some topics are clear others need improvement
22	Data Structures	Prof. Sharma	2	3	1	3	3	Yes	Some topics are clear others need improvement
23	Operating Systems	Prof. Mehta	4	5	4	5	5	Yes	Great examples and supportive teaching
24	Computer Networks	Dr. Joshi	3	1	4	3	3	Yes	Decent teaching overall
25	Operating Systems	Prof. Mehta	4	4	2	3	1	No	Needs improvement in teaching methods
26	Data Structures	Prof. Sharma	4	2	2	1	4	Yes	Teaching is average but understandable
27	Computer Networks	Dr. Joshi	4	4	1	3	4	Yes	Some topics are clear others need improvement
28	Machine Learning	Dr. Patil	4	4	1	1	3	No	Teaching is average but understandable
29	Machine Learning	Dr. Patil	5	3	3	5	4	Yes	Great examples and supportive teaching
30	Machine Learning	Dr. Patil	4	4	4	2	4	Yes	Pace is okay but can be better
31	Data Structures	Prof. Sharma	4	5	5	5	4	Yes	Very helpful and well structured classes

... kpis cleaned TABLE **Student Feedback 1000 Updated**

- **Figure A2: Cleaned Dataset Screenshot (after preprocessing)** *(Snipping Tool capture — Excel sheet after applying formulas and preprocessing steps, duplicates removed, standardized values, and issue categories added.)*

Course_Name	Faculty_Name	Clarity	Engagement	Pace	Study_Materials	Overall_Rating	Would_Recommend	Feedback_Text	Issue_Category.
Operating Systems	Prof. Mehta	4	4	1	4	3	Yes	Some topics are clear others need improvement	Other
Data Structures	Prof. Sharma	2	3	1	3	3	Yes	Some topics are clear others need improvement	Other
Operating Systems	Prof. Mehta	4	5	4	5	5	Yes	Great examples and supportive teaching	Positive Feedback
Computer Networks	Dr. Joshi	3	1	4	3	3	Yes	Decent teaching overall	Neutral Feedback
Operating Systems	Prof. Mehta	4	4	2	3	1	No	Needs improvement in teaching methods	Teaching Quality Issue
Data Structures	Prof. Sharma	4	2	2	1	4	Yes	Teaching is average but understandable	Other
Computer Networks	Dr. Joshi	4	4	1	3	4	Yes	Some topics are clear others need improvement	Other
Machine Learning	Dr. Patil	4	4	1	1	3	No	Teaching is average but understandable	Other
Machine Learning	Dr. Patil	5	3	3	5	4	Yes	Great examples and supportive teaching	Positive Feedback
Machine Learning	Dr. Patil	4	4	4	2	4	Yes	Pace is okay but can be better	Other
Data Structures	Prof. Sharma	4	5	5	5	4	Yes	Very helpful and well structured classes	Other
Computer Networks	Dr. Joshi	3	5	4	4	4	Yes	Very helpful and well structured classes	Other
Data Structures	Prof. Sharma	4	4	3	2	3	No	Needs improvement in teaching methods	Teaching Quality Issue
Data Structures	Prof. Sharma	3	3	3	5	4	Yes	Engaging lectures and easy to understand	Positive Feedback
Operating Systems	Prof. Mehta	5	5	5	4	5	Yes	Very helpful and well structured classes	Other
Machine Learning	Dr. Patil	1	4	1	3	4	Yes	Pace is okay but can be better	Other
Computer Networks	Dr. Joshi	5	3	3	5	5	Yes	Great examples and supportive teaching	Positive Feedback
Data Structures	Prof. Sharma	3	3	3	4	5	Yes	Very helpful and well structured classes	Other
Machine Learning	Dr. Patil	3	3	3	5	5	Yes	Great examples and supportive teaching	Positive Feedback
Data Structures	Prof. Sharma	1	2	3	1	4	No	Teaching is average but understandable	Other
Machine Learning	Dr. Patil	1	1	3	3	4	No	Teaching is average but understandable	Other
Dbms	Dr. Kulkarni	5	5	4	4	5	Yes	Engaging lectures and easy to understand	Positive Feedback
Machine Learning	Dr. Patil	4	3	5	3	5	Yes	Great examples and supportive teaching	Positive Feedback
Dbms	Dr. Kulkarni	3	3	3	3	3	No	Pace is okay but can be better	Other
Dbms	Dr. Kulkarni	2	4	3	2	1	No	Too fast and confusing teaching style	Pace Issue
Dbms	Dr. Kulkarni	3	3	5	4	5	Yes	Great examples and supportive teaching	Positive Feedback
Dbms	Dr. Kulkarni	1	1	1	2	2	No	Lectures are boring and hard to follow	Engagement Issue
Machine Learning	Dr. Patil	2	2	4	1	2	No	Lectures are boring and hard to follow	Engagement Issue
Data Structures	Prof. Sharma	4	2	3	1	4	Yes	Decent teaching overall	Neutral Feedback
Operating Systems	Prof. Mehta	1	2	1	3	4	No	Teaching is average but understandable	Other

pivot_tables | Dashboard | kpis | cleaned TABLE | Student Feedback 1000 Updated

- **Figure A3: Excel Automated Workflow (Cleaning Stage)** *(Snipping Tool capture — Excel interface showing formulas, conditional formatting, and automated categorization steps used for cleaning instead of Power BI Query Editor.)*

K2

```
=IF(ISNUMBER(SEARCH("boring",J2)),"Engagement Issue",
IF(ISNUMBER(SEARCH("confusing",J2)),"Pace Issue",
IF(ISNUMBER(SEARCH("not clearly explained",J2)),"Clarity Issue",
IF(ISNUMBER(SEARCH("Needs improvement",J2)),"Teaching Quality Issue",
IF(ISNUMBER(SEARCH("Great examples",J2)),"Positive Feedback",
IF(ISNUMBER(SEARCH("Engaging lectures",J2)),"Positive Feedback",
IF(ISNUMBER(SEARCH("Decent teaching",J2)),"Neutral Feedback","Other")))))))
```

	E	F	G	H	I	J	K
1	Engagement	Pace	Study_Materials	Overall_Rating	Would_Recommend	Feedback_Text	Issue_Category.
2	2	2	2	2	No	Concepts are not clearly explained	Clarity Issue
3	4	3	2	1	No	Lectures are boring and hard to follow	Engagement Issue
4	4	5	3	4	Yes	Great examples and supportive teaching	Positive Feedback
5	4	3	4	3	No	Lectures are boring and hard to follow	Engagement Issue
6	1	2	2	3	No	Too fast and confusing teaching style	Pace Issue
7	2	4	1	4	No	Decent teaching overall	Neutral Feedback
8	4	4	2	2	No	Lectures are boring and hard to follow	Engagement Issue
9	3	4	1	1	No	Concepts are not clearly explained	Clarity Issue
10	2	1	3	3	Yes	Some topics are clear others need improvement	Other
11	3	1	4	2	No	Too fast and confusing teaching style	Pace Issue
12	5	5	4	5	Yes	Excellent teaching and very clear explanations	Other
13	4	2	1	3	No	Too fast and confusing teaching style	Pace Issue
14	4	4	1	4	No	Some topics are clear others need improvement	Other
15	4	1	2	1	No	Concepts are not clearly explained	Clarity Issue
16	1	4	3	4	Yes	Pace is okay but can be better	Other
17	3	2	2	3	No	Some topics are clear others need improvement	Other
18	1	3	3	2	No	Lectures are boring and hard to follow	Engagement Issue
19	1	1	4	3	No	Concepts are not clearly explained	Clarity Issue
20	2	3	3	3	No	Concepts are not clearly explained	Clarity Issue
21	1	4	2	3	No	Too fast and confusing teaching style	Pace Issue
22	4	1	4	3	Yes	Some topics are clear others need improvement	Other
23	3	1	3	3	Yes	Some topics are clear others need improvement	Other

pivot_tables | Dashboard | kpis | cleaned TABLE | Student Feedback 1000 Updated

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- **Figure A4: PivotTable (Faculty Comparison)** *(Snipping Tool capture — Excel PivotTable summarizing average ratings per faculty, recommendation percentages, category distribution, clarity Vs engagement and overall dashboard summary. This figure highlights the tabular analysis prepared on a separate sheet.)*

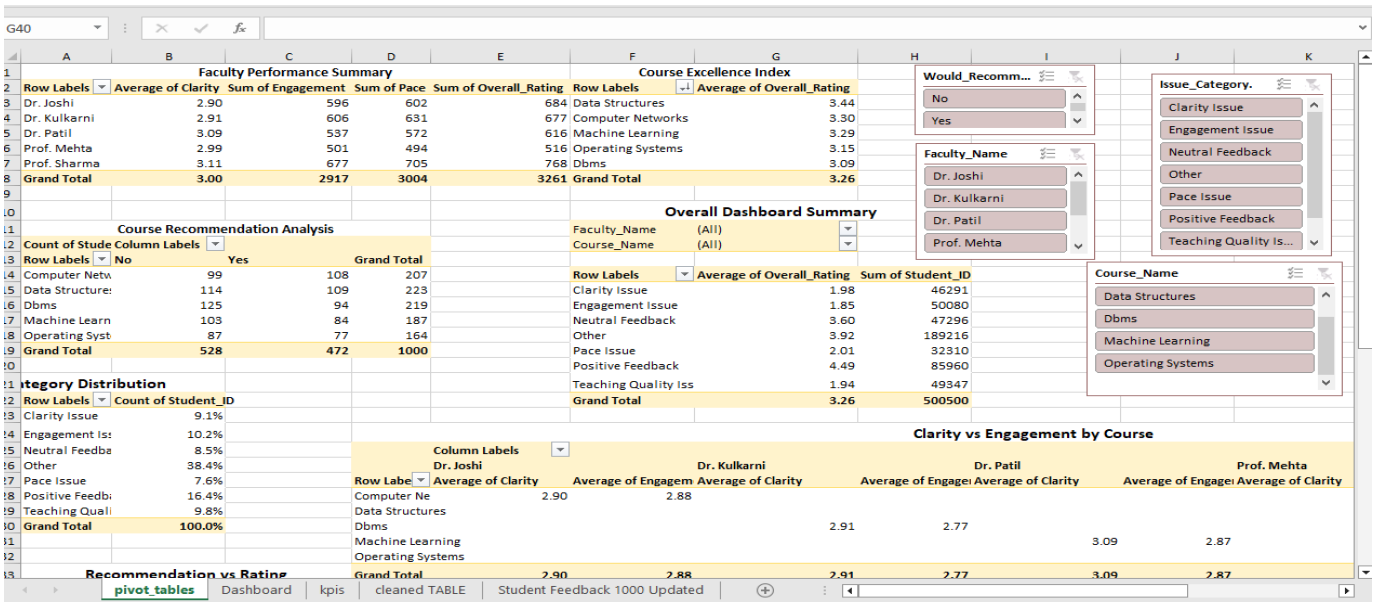


Figure A5: Dashboard (Multiple Visuals) *(Snipping Tool capture — Excel dashboard sheet showing faculty comparison charts, recommendation analysis, top performing, overall rating and satisfaction KPIs. This figure demonstrates how PivotTable outputs were converted into a comprehensive dashboard for easier interpretation.)*

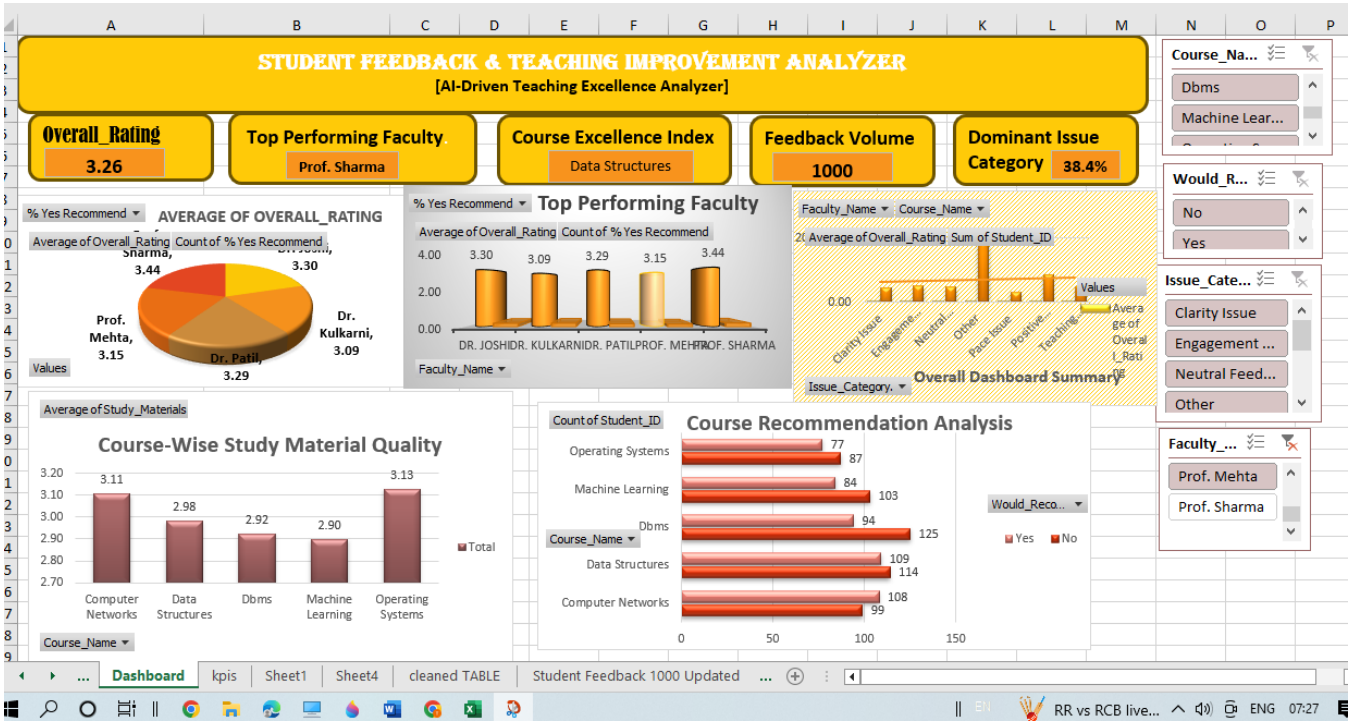
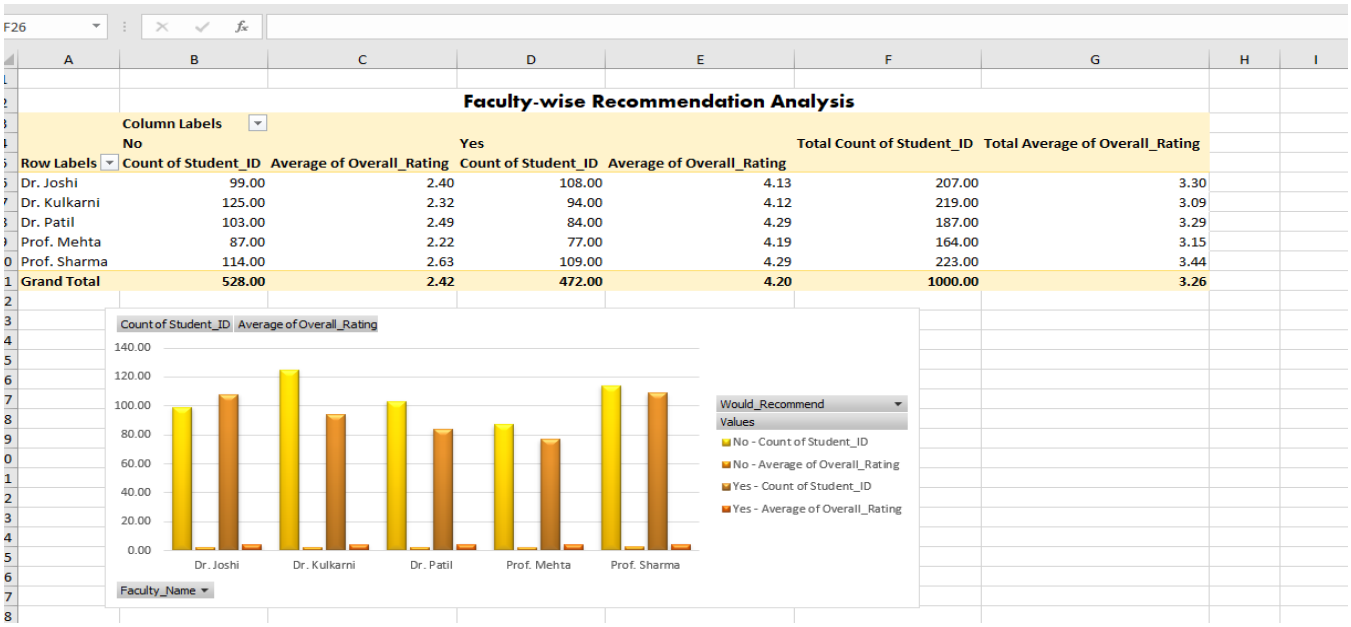


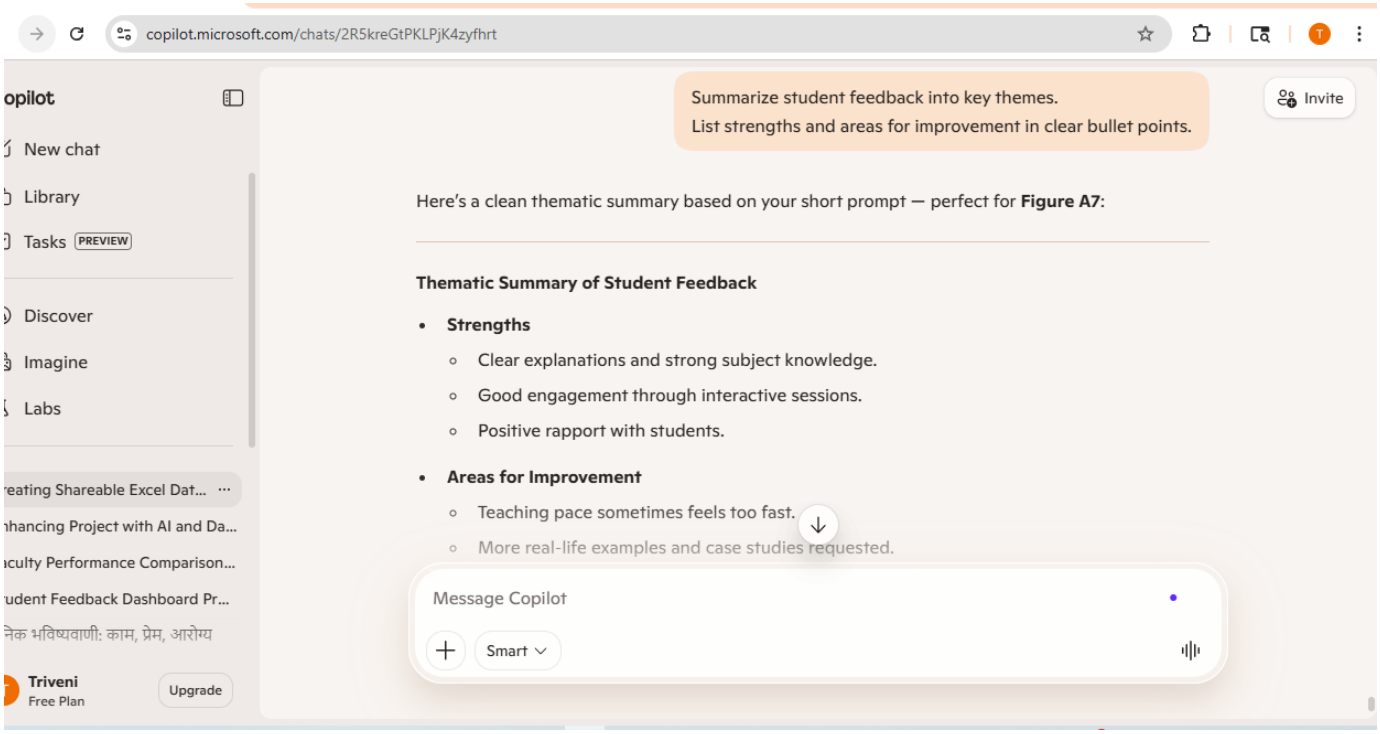
Figure A6: Faculty-wise Recommendation Analysis (Table & Chart) *(Snipping Tool capture — PivotTable and chart together, showing recommendation counts and ratings across faculty members.)*



Faculty-wise Recommendation Analysis

Row Labels	No: Count of Student_ID	No: Average of Overall_Rating	Yes: Count of Student_ID	Yes: Average of Overall_Rating	Total Count of Student_ID	Total Average of Overall_Rating
Dr. Joshi	99.00	2.40	108.00	4.13	207.00	3.30
Dr. Kulkarni	125.00	2.32	94.00	4.12	219.00	3.09
Dr. Patil	103.00	2.49	84.00	4.29	187.00	3.29
Prof. Mehta	87.00	2.22	77.00	4.19	164.00	3.15
Prof. Sharma	114.00	2.63	109.00	4.29	223.00	3.44
Grand Total	**528.00**	**2.42**	**472.00**	**4.20**	**1000.00**	**3.26**

- **Figure A7: Copilot Summary Screenshot** *(Snipping Tool capture — Copilot-generated thematic summary of qualitative feedback.)*



Appendix E — Interpretation Notes

Figure A1: Dataset Snapshot This figure shows the raw student feedback dataset structure, including student IDs, ratings, and recommendation fields. It serves as the foundation for subsequent analysis.

Figure A2: Initial Pivot Setup This figure presents the first pivot table arrangement, demonstrating how feedback data was organized for analysis. It documents the workflow rather than providing insights.

Figure A3: Basic Feedback Summary This figure highlights the overall distribution of ratings and recommendations in the dataset. It provides a descriptive overview but does not require detailed interpretation.

Figure A4: Faculty Comparison Pivot This table highlights the variation in average ratings across faculty members. It shows clear differences in teaching effectiveness, with some faculty consistently scoring higher in clarity and engagement.

Figure A5: Dashboard Overview The dashboard integrates multiple visuals — faculty comparison, recommendation analysis, issue frequency, and satisfaction KPIs. It provides a holistic view of teaching performance and student sentiment, making patterns easier to interpret at a glance.

Figure A6: Faculty-wise Recommendation Analysis This figure shows that students who recommended faculty gave significantly higher ratings (around 4.1–4.3) compared to those who did not (around 2.2–2.6). Prof. Sharma has the highest overall average rating (3.44), while Dr. Kulkarni has the lowest (3.09). This highlights clear differences in faculty performance and student satisfaction.

Figure A7: Copilot/Chatbot Summary The AI-generated thematic summary of qualitative feedback reveals that students value clarity, subject knowledge, and engagement. However, concerns remain about pacing, the need for more real-life examples, and greater individual attention. This complements the quantitative analysis by adding context from open-ended responses.